

Backend Test Report

Small Business Operations Platform

Project	Small Business Operations Platform
Team	Claudius (Group 61)
Institution	IITM Online BS Degree Program, IIT Madras
Domain	Small Retail Clothing Stores (Tier 2/3/4)
Testing Framework	Jest 29 + ts-jest + Supertest
Database	PostgreSQL 16 (smallbiz_test)
Report Date	April 2026

480 / 480

Tests Passing across 14 Suites

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1. Executive Summary

This report documents the complete backend integration test suite for the Small Business Operations Platform. All 480 tests run as real HTTP calls against a dedicated PostgreSQL test database using Jest 29 and Supertest. Tests were written using test-driven development, which directly identified 7 critical defects before any manual QA was performed.

480 TOTAL TESTS	480 TESTS PASSED	14 TEST SUITES	33 API ENDPOINTS	7 BUGS FIXED
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1.1 Test Suite Breakdown

Test Suite	File	Tests	Coverage Area	Status
Authentication	<code>auth.test.ts</code>	27	Register, login, JWT validation	ALL PASS
User Management	<code>users.test.ts</code>	67	CRUD, passwords, PINs, avatars, RBAC	ALL PASS
Store Configuration	<code>store.test.ts</code>	41	Profile, billing mode, retention, GPS	ALL PASS
Categories	<code>categories.test.ts</code>	23	Category CRUD, uniqueness constraints	ALL PASS
Price Bands	<code>priceBands.test.ts</code>	32	Price band CRUD, stock queries	ALL PASS
Inventory Batches	<code>inventory.test.ts</code>	62	Batches, matrix, adjust, defective, close	ALL PASS
Sales	<code>sales.test.ts</code>	22	FIFO deduction, customers, custom items	ALL PASS
Returns and Exchanges	<code>returns.test.ts</code>	30	Returns, exchanges, stock restoration	ALL PASS

Test Suite	File	Tests	Coverage Area	Status
Vendors	<code>vendors.test.ts</code>	45	Vendor directory, soft-delete, suggest-order	ALL PASS
Reorders	<code>reorders.test.ts</code>	38	Create reorder, status updates, message text	ALL PASS
Attendance and GPS	<code>attendance.test.ts</code>	20	Clock-in/out, GPS geofencing, roster	ALL PASS
Staff Leaderboard	<code>staff.test.ts</code>	12	Leaderboard rankings, RBAC, sort order	ALL PASS
Dashboard	<code>dashboard.test.ts</code>	15	KPI summary, daily charts, RBAC	ALL PASS
Audit Log and Wipe	<code>audit.test.ts</code>	19	Audit log, ephemeral data wipe	ALL PASS
Total		480		480 PASS

2. Test Cases - Authentication

2.1 POST /api/auth/register

ID	Test Description	Input Condition	Expected Result	Status
TC-01	Create store and owner account	Valid shopName, ownerName, email, phone, password, address	201 + token + user object	PASS
TC-02	Accept optional logo field	Valid body with logo URL	201 + token	PASS
TC-03	Reject duplicate email	Email already registered in database	400 error	PASS
TC-04	Reject duplicate shop name	shopName already in use	400 error	PASS
TC-05	Reject missing shopName	Body without shopName field	400	PASS
TC-06	Reject missing ownerName	Body without ownerName field	400	PASS
TC-07	Reject missing email	Body without email field	400	PASS
TC-08	Reject invalid email format	email: "notanemail"	400	PASS
TC-09	Reject phone not 10 digits	phone: "12345"	400	PASS
TC-10	Reject phone with non-digit characters	phone: "999999999a"	400	PASS
TC-11	Reject password shorter than 6 characters	password: "abc"	400	PASS
TC-12	Reject missing address	Body without address field	400	PASS
TC-13	Reject empty body	{ }	400	PASS

2.2 POST /api/auth/login

ID	Test Description	Input Condition	Expected Result	Status
TC-14	Owner login with email and password	Correct credentials	200 + JWT token	PASS
TC-15	Manager login with email and password	Correct credentials	200 + JWT token	PASS
TC-16	Staff login with phone and PIN	Correct phone + 4-digit PIN	200 + JWT token	PASS
TC-17	Verify token is a valid JWT	Any successful login response	3-part dot-separated string	PASS
TC-18	Reject wrong password	Correct email, wrong password	401	PASS
TC-19	Reject wrong PIN	Correct phone, wrong PIN	401	PASS
TC-20	Reject non-existent email	Unknown email address	401	PASS
TC-21	Reject non-existent phone	Unknown phone number	401	PASS
TC-22	Reject missing identifier	No email or phone in body	400	PASS
TC-23	Reject missing credential	Email present, no password	400	PASS
TC-24	Reject empty body	<code>{}</code>	400	PASS
TC-25	Reject empty string identifier	<code>email: ""</code>	400	PASS
TC-26	Reject empty string credential	<code>password: ""</code>	400	PASS
TC-27	Staff login via email fails gracefully	Staff user has no password_hash set	401	PASS

3. Test Cases - User Management

ID	Test Description	Role Tested	Expected Result	Status
TC-28	List users returns array with correct shape	Owner	200 + array with id, name, phone, role, isActive	PASS
TC-29	Manager can list users	Manager	200 + array	PASS
TC-30	Staff can list users	Staff	200 + array	PASS
TC-31	No cross-store user data leakage	Owner (different store)	Only own-store users returned	PASS
TC-32	Owner adds staff user with phone and PIN	Owner	201 + user object	PASS
TC-33	Owner adds manager user with email and password	Owner	201 + user object	PASS
TC-34	Manager can add staff	Manager	201	PASS
TC-35	Staff cannot add users	Staff	403	PASS
TC-36	Reject duplicate phone number	Owner	409	PASS
TC-37	Reject staff PIN that is not 4 digits	Owner	400	PASS
TC-38	Reject manager creation without password	Owner	400	PASS
TC-39	Owner soft-deletes staff user	Owner	200 + user absent from GET response	PASS
TC-40	Owner cannot delete themselves	Owner	403 protected	PASS
TC-41	Manager cannot delete users	Manager	403	PASS
TC-42	Reject cross-store update attempt	Owner (different store)	404	PASS
TC-43		Owner		PASS

ID	Test Description	Role Tested	Expected Result	Status
	Password change requires correct current password		401 if current password is wrong	
TC-44	Enforce minimum 6-character new password	Owner	400 if too short	PASS
TC-45	PIN change only valid for staff role	Owner	400 if applied to manager or owner	PASS
TC-46	Staff can update their own avatar	Staff	200	PASS
TC-47	Staff cannot update another user avatar	Staff	403	PASS
TC-48	Unauthenticated requests rejected on all user routes	No token	401	PASS

4. Test Cases - Store Configuration

ID	Test Description	Input Condition	Expected Result	Status
TC-49	Owner updates store name and address	Valid patch body	200 + updated store object	PASS
TC-50	Manager cannot update store profile	Manager token	403	PASS
TC-51	Owner switches billing mode to ephemeral	<code>billingMode: "ephemeral"</code>	200	PASS
TC-52	Owner switches billing mode to structured	<code>billingMode: "structured"</code>	200	PASS
TC-53	Reject invalid billing mode value	<code>billingMode: "unknown"</code>	400	PASS
TC-54	Owner sets retention period in days	Valid retention value	200	PASS
TC-55	Reject retention period below 1 day	<code>retentionDays: 0</code>	400	PASS

ID	Test Description	Input Condition	Expected Result	Status
TC-56	Owner sets GPS coordinates and geofence radius	Valid lat, lng, radiusM	200 + persisted values	PASS
TC-57	gpsRequireClockIn = true persisted to database	gpsRequireClockIn: true	200 + confirmed in DB row	PASS
TC-58	gpsRequireClockOut = true persisted to database	gpsRequireClockOut: true	200 + confirmed in DB row	PASS
TC-59	Reject zero GPS radius	gpsRadiusM: 0	400	PASS
TC-60	Reject string values for GPS coordinates	gpsLatitude: "abc"	400	PASS
TC-61	Manager cannot update GPS settings	Manager token	403	PASS
TC-62	Staff cannot update GPS settings	Staff token	403	PASS
TC-63	Unauthenticated request rejected	No token	401	PASS

5. Test Cases - Categories and Price Bands

5.1 Categories

ID	Test Description	Input Condition	Expected Result	Status
TC-64	Owner lists categories including seeded data	Owner token	200 + array	PASS
TC-65	Category response shape is correct	Owner token	id, name, storeId, createdAt present	PASS
TC-66	No cross-store category leakage	Owner (different store)	Only own-store categories returned	PASS
TC-67	Owner creates category	Valid name	201 + created object	PASS
TC-68	Manager creates category	Manager token, valid name	201	PASS
TC-69	Staff cannot create category	Staff token	403	PASS
TC-70	Reject empty category name	name: ""	400	PASS
TC-71	Reject duplicate category name in same store	Name already used in store	409	PASS
TC-72	New category persists in subsequent GET	POST then GET	Category visible in list	PASS
TC-73	Category with active price bands is protected from deletion	Delete category with bands	Category still exists after attempt	PASS

5.2 Price Bands

ID	Test Description	Input Condition	Expected Result	Status
TC-74	Owner creates price band under category	Valid categoryId and price	201 + band object	PASS
TC-75	Reject zero or negative price	price: 0	400	PASS
TC-76		Foreign categoryId	404	PASS

ID	Test Description	Input Condition	Expected Result	Status
	Reject categoryId from another store			
TC-77	GET price bands with stock returns correct totals	Owner token	200 + totalStock per band	PASS
TC-78	Staff can read price bands	Staff token	200	PASS
TC-79	Staff cannot create price bands	Staff token	403	PASS

6. Test Cases - Inventory Batches

ID	Test Description	Input Condition	Expected Result	Status
TC-80	Owner adds batch of 20 units at price band	Valid priceBandId and quantityAdded	201 + quantityRemaining equals quantityAdded	PASS
TC-81	Manager adds batch	Manager token	201	PASS
TC-82	Staff cannot add batch	Staff token	403	PASS
TC-83	Batch without optional fields succeeds	No vendorId, costPrice, or notes	201 with nullable optional fields	PASS
TC-84	Reject quantityAdded of zero	quantityAdded: 0	400	PASS
TC-85	Reject negative quantityAdded	quantityAdded: -5	400	PASS
TC-86	Reject float quantityAdded	quantityAdded: 1.5	400 (integer required)	PASS
TC-87	Reject priceBandId from another store	Foreign priceBandId	404	PASS
TC-88	Reject zero or negative costPrice	costPrice: 0	400	PASS

ID	Test Description	Input Condition	Expected Result	Status
TC-89	Newly added batch appears in GET response	POST then GET	Batch visible in list	PASS
TC-90	Filter low_stock returns batches with qty below 10	?filter=low_stock	200 filtered array	PASS
TC-91	Filter aging returns batches older than 60 days	?filter=aging	200 (empty for fresh test data)	PASS
TC-92	Filter new_arrivals returns last 7 days	?filter=new_arrivals	200 including seeded batch	PASS
TC-93	Reject invalid filter value	?filter=invalid	400	PASS
TC-94	Margin computed when costPrice and price are present	Batch with costPrice	margin field not null	PASS
TC-95	Adjust quantity upward	Action adjust, positive quantity	200 + increased quantity	PASS
TC-96	Adjust quantity downward	Action adjust, lower quantity	200 + decreased quantity	PASS
TC-97	Adjust to zero depletes batch	Action adjust, quantity 0	200 + quantity is 0	PASS
TC-98	Reject negative quantity for adjust action	Action adjust, quantity -1	400	PASS
TC-99	Staff cannot adjust batch	Staff token	403	PASS
TC-100	Adjustment reflected in subsequent GET	PATCH then GET	Updated quantity in response	PASS
TC-101	Close batch sets quantityRemaining to zero	Action close	200 + quantityRemaining is 0	PASS

ID	Test Description	Input Condition	Expected Result	Status
TC-102	Staff cannot close batch	Staff token	403	PASS
TC-103	Mark batch defective appends [DEFECTIVE] to notes	Action defective	200 + notes contains [DEFECTIVE]	PASS
TC-104	Defective flag appends to existing notes text	Batch with existing notes	[DEFECTIVE] added to existing text	PASS
TC-105	Defective flag on null notes sets notes to [DEFECTIVE]	Batch with null notes	notes becomes "[DEFECTIVE]"	PASS
TC-106	Defective action does not change quantityRemaining	Action defective	Quantity unchanged after action	PASS
TC-107	Reject invalid action value	action: "unknown"	400	PASS
TC-108	Non-existent batch ID returns 404	Invalid batch ID	404	PASS
TC-109	Stock matrix has correct category and band structure	Owner token	200 + band "299" shows stock 20	PASS
TC-110	Stock matrix values are non-negative integers	Owner token	All values greater than or equal to 0	PASS

7. Test Cases - Sales and Returns

7.1 Sales

ID	Test Description	Input Condition	Expected Result	Status
TC-111	Staff creates a sale with id and createdAt in response	Valid sale body	201 + id + createdAt	PASS
TC-112	Owner creates sale with UPI payment method	<code>paymentMethod: "upi"</code>	201	PASS
TC-113	Manager creates sale with store credit payment	<code>paymentMethod: "store_credit"</code>	201	PASS
TC-114	FIFO deducts from oldest batch first then spills to newer	Quantity exceeds oldest batch	201 + oldest batch depleted first	PASS
TC-115	Insufficient stock returns descriptive error with item name	Quantity exceeds all stock	400 with item name in error	PASS
TC-116	priceBandId with zero batches reports insufficient stock	Band exists but has no batches	400	PASS
TC-117	Custom item with priceBandId 0 triggers no stock deduction	<code>priceBandId: 0</code>	201 + stock unchanged	PASS
TC-118	Custom item mixed with normal item deducts only normal item	Mixed items array	201 + only normal stock deducted	PASS
TC-119	New customer record created when phone first seen	New customerPhone	201 + customer record exists in DB	PASS
TC-120	Existing customer name updated on second sale with same phone	Same phone, different name	201 + name updated	PASS

ID	Test Description	Input Condition	Expected Result	Status
TC-121	Sale without customerPhone creates no customer record	No customerPhone field	201 + no customer inserted	PASS
TC-122	discountAmount defaults to zero when omitted	No discountAmount in body	201 + discountAmount is 0	PASS
TC-123	Reject negative discountAmount	<code>discountAmount: -10</code>	400	PASS
TC-124	Reject invalid paymentMethod	<code>paymentMethod: "crypto"</code>	400	PASS
TC-125	Reject empty items array	<code>items: []</code>	400	PASS
TC-126	Reject item missing categoryName	Item without categoryName	400	PASS
TC-127	Reject item quantity less than 1	<code>quantity: 0</code>	400	PASS
TC-128	Reject non-10-digit customerPhone	<code>customerPhone: "123"</code>	400	PASS
TC-129	Reject unauthenticated request	No token	401	PASS

7.2 Returns and Exchanges

ID	Test Description	Input Condition	Expected Result	Status
TC-130	Owner creates return for existing sale	Valid saleId and items	201	PASS
TC-131	Manager creates return	Manager token	201	PASS
TC-132	Staff creates return (no role restriction on POST)	Staff token	201	PASS
TC-133	Exchange type is accepted	<code>type: "exchange"</code>	201	PASS

ID	Test Description	Input Condition	Expected Result	Status
TC-134	Full quantity return accepted	Return all sold units	201	PASS
TC-135	Stock restored to inventory after return	Partial return	quantityRemaining increases by returned qty	PASS
TC-136	Full return fully restores stock to original level	Return all units	Stock back to original level	PASS
TC-137	Reject return quantity exceeding sold quantity	Return qty greater than sale qty	400	PASS
TC-138	Non-existent saleId returns 404	Invalid saleId	404	PASS
TC-139	Non-existent saleItemId returns 404	Invalid saleItemId	404	PASS
TC-140	Returns list ordered newest first	GET /api/returns	createdAt descending order	PASS
TC-141	Staff cannot list returns	Staff token	403	PASS
TC-142	Reject missing type field	Body without type	400	PASS
TC-143	Reject invalid type value	type: "unknown"	400	PASS
TC-144	Reject quantity of zero	quantity: 0	400	PASS

8. Test Cases - Vendors and Reorders

8.1 Vendors

ID	Test Description	Input Condition	Expected Result	Status
TC-145	Owner creates vendor with name, phone, city	Full vendor body	201 + vendor object	PASS
TC-146		Name only		PASS

ID	Test Description	Input Condition	Expected Result	Status
	Vendor with name only (all optional fields omitted)		201 with nullable optional fields	
TC-147	Manager creates vendor	Manager token	201	PASS
TC-148	Staff cannot create vendor	Staff token	403	PASS
TC-149	Reject vendor name shorter than 2 characters	name: "A"	400	PASS
TC-150	Two vendors with same phone number allowed	Duplicate phone	201 (no unique constraint on vendor phone)	PASS
TC-151	Owner updates vendor name	Valid patch body	200 + updated vendor	PASS
TC-152	Manager updates vendor	Manager token	200	PASS
TC-153	Staff cannot update vendor	Staff token	403	PASS
TC-154	Owner soft-deletes vendor	Owner token	200 + vendor absent from GET	PASS
TC-155	Deleting an already-deleted vendor returns 404	Second delete of same vendor	404 not 200	PASS
TC-156	Manager cannot delete vendor	Manager token	403	PASS
TC-157	Suggest-order returns per-band suggested quantities	GET vendors/:id/suggest-order	200 + suggestedQty array	PASS
TC-158	suggestedQty uses formula: $\max(0, \text{ceil}(\text{dailyAvg} \times 14) - \text{currentStock})$	Vendor with sales history	Correct computed value	PASS
TC-159	Suggest-order for non-existent vendor returns 404	Invalid vendorId	404	PASS

8.2 Reorders

ID	Test Description	Input Condition	Expected Result	Status
TC-160	Owner creates reorder with vendorId and items	Valid reorder body	201 + reorder object	PASS

ID	Test Description	Input Condition	Expected Result	Status
TC-161	Manager creates reorder	Manager token	201	PASS
TC-162	Staff cannot create reorder	Staff token	403	PASS
TC-163	messageText generated on creation with vendor name and items	Valid reorder	Non-empty formatted string	PASS
TC-164	Items with finalQty zero are excluded from messageText	Item with finalQty 0	Zero-qty items absent from text	PASS
TC-165	messageText preserved and accessible via GET	GET /api/reorders after creation	Same text available later	PASS
TC-166	Reorder status updated to acknowledged	status: "acknowledged"	200 + updated status	PASS
TC-167	Reorder status updated to fulfilled	status: "fulfilled"	200 + updated status	PASS
TC-168	Reorder status updated to draft	status: "draft"	200 + updated status	PASS
TC-169	Reject invalid status value	status: "shipped"	400	PASS
TC-170	Non-existent reorder ID returns 404	Invalid reorderId	404	PASS
TC-171	Non-existent vendorId returns 404	Foreign vendorId	404	PASS
TC-172	Reorders list ordered by createdAt descending	GET /api/reorders	Newest reorder appears first	PASS

9. Test Cases - Attendance and GPS Geofencing

ID	Test Description	Input Condition	Expected Result	Status
TC-173	Staff clocks in successfully	Valid lat and lng	200 + clockedIn: true + checkInAt	PASS
TC-174	Reject duplicate clock-in on same day	Staff already clocked in today	400 already clocked in error	PASS
TC-175	Owner clocks in without GPS requirement	GPS not required by default	200	PASS
TC-176	GPS required with valid coords within radius succeeds	gpsRequireClockIn true, coords inside	200	PASS
TC-177	GPS required with coords outside radius is rejected	gpsRequireClockIn true, coords outside	400 geofence error	PASS
TC-178	GPS not required accepts coords outside radius	gpsRequireClockIn false, coords outside	200	PASS
TC-179	Reject missing lat and lng fields	Body without coordinates	400	PASS
TC-180	Staff clocks out successfully after clock-in	Valid lat and lng after clock-in	200 + clockedIn: false + checkOutAt	PASS
TC-181	Reject clock-out without prior clock-in	No clock-in record for today	400	PASS
TC-182	Reject second clock-out after already clocked out	Already has checkOutAt value	400	PASS
TC-183	GPS not required allows clock-out from remote location	gpsRequireClockOut false	200	PASS
TC-184		Clock-in then clock-out		PASS

ID	Test Description	Input Condition	Expected Result	Status
	Attendance DB row records lat and lng coordinates		check_in_lat and check_out_lat saved correctly	
TC-185	Staff cannot view today roster	Staff token	403	PASS
TC-186	Owner can view today roster	Owner token	200 + array	PASS
TC-187	Manager can view today roster	Manager token	200	PASS
TC-188	Manager can force clock-out a staff member	Manager token + userId	200	PASS
TC-189	Unauthenticated clock-in rejected	No token	401	PASS
TC-190	Unauthenticated clock-out rejected	No token	401	PASS

10. Test Cases - Dashboard, Staff Leaderboard, and Audit

10.1 Dashboard

ID	Test Description	Input Condition	Expected Result	Status
TC-191	Owner receives KPI summary with all required fields	Owner token	200 + KPI object	PASS
TC-192	Manager receives KPI summary	Manager token	200	PASS
TC-193	activeVendors count includes seeded vendor	Owner token	count is 1 or more	PASS
TC-194	Staff cannot access dashboard	Staff token	403	PASS
TC-195		?range=weekly	200 + array of rows	PASS

ID	Test Description	Input Condition	Expected Result	Status
	Daily summary with range=weekly returns array			
TC-196	Daily summary with range=monthly returns array	? range=monthly	200 + array of rows	PASS
TC-197	Daily summary rows have correct shape	Owner token	bandPrice, totalQtySold, totalRevenue present	PASS
TC-198	Reject invalid range value	?range=yearly	400	PASS

10.2 Staff Leaderboard

ID	Test Description	Input Condition	Expected Result	Status
TC-199	Owner receives leaderboard array	Owner token	200 + array	PASS
TC-200	All active seeded users appear in leaderboard	Owner token	3 entries for owner, manager, staff	PASS
TC-201	Leaderboard sorted by revenueToday descending	Owner token	Correct descending sort order	PASS
TC-202	salesCountToday defaults to zero when no sales recorded	No sales today	salesCountToday is 0	PASS
TC-203	Staff can access leaderboard	Staff token	200	PASS
TC-204	Only active users from same store included	Owner token	No cross-store or inactive entries	PASS

10.3 Audit Log and Ephemeral Wipe

ID	Test Description	Input Condition	Expected Result	Status
TC-205	Owner views audit log (empty on fresh database)	Owner token	200 + empty array	PASS
TC-206		Manager token	403	PASS

ID	Test Description	Input Condition	Expected Result	Status
	Manager cannot view audit log			
TC-207	Staff cannot view audit log	Staff token	403	PASS
TC-208	Wipe with correct password prunes ephemeral sales	Correct owner password	200 + recordsPruned greater than 0	PASS
TC-209	Wipe with no ephemeral sales reports zero records pruned	No ephemeral sales in DB	200 + recordsPruned is 0	PASS
TC-210	Reject wipe with wrong password	Wrong password	401	PASS
TC-211	Wipe entry appears in audit log afterwards	GET /audit/log after wipe	Log entry with type and timestamp	PASS
TC-212	Manager cannot trigger wipe	Manager token	403	PASS
TC-213	Staff cannot trigger wipe	Staff token	403	PASS
TC-214	Reject wipe with missing password field	Body without password	400	PASS

11. Bugs Found and Fixed During Testing

The test suite identified 7 critical defects through automated integration testing. All bugs were resolved before delivery. Each fix is documented with its root cause and the specific change applied.

BUG-01 - GPS Geofencing Completely Non-Functional (Story 7.1)

`frontend/src/pages/StaffPage.tsx | backend/src/services/attendance.ts`

Symptom: Enabling "Require GPS on Clock-In" had no effect. Staff could clock in from any location regardless of the toggle state.

Root Cause: The GPS enforcement flags were stored only in Zustand (browser memory) and were never written to the database. The backend attendance service always read from an empty `gps_settings` table and therefore never enforced any policy on any request.

FIX-01

Added `gps_require_clock_in` and `gps_require_clock_out` boolean columns to the `stores` table. Updated `PATCH /store/gps-settings` to accept and persist both fields. The backend attendance service now reads these flags directly from the store row on every clock-in and clock-out request. The frontend `StaffPage` now calls the API on every toggle change to persist the new value immediately.

BUG-02 - Returns Service: column "store_id" does not exist

`backend/src/services/returns.ts`

Symptom: `POST /api/returns` always failed with a PostgreSQL error: column "store_id" does not exist.

Root Cause: The SQL query attempted to select `store_id` from the `sale_items` table, which has no such column. The table only contains line-item data and does not carry a store reference.

FIX-02

Changed the query to pass `storeId` as the `$3` bind parameter rather than attempting to derive it from the `sale_items` table. The value is already available from the authenticated request context.

BUG-03 - Returns Service: Inconsistent Types for Parameter \$1

`backend/src/services/returns.ts`

Symptom: PostgreSQL rejected the refund calculation query with: inconsistent types deduced for parameter \$1.

Root Cause: The same bind parameter `$1` appeared in both an integer comparison in the WHERE clause and in arithmetic multiplication, causing the query planner to fail on type inference since the parameter cannot be both integer and numeric simultaneously.

FIX-03

Added explicit PostgreSQL type casts: `$1::int` where an integer comparison is needed and `$1::numeric` where multiplication is performed. This resolves the ambiguity and allows the query planner to proceed correctly.

BUG-04 - Vendor Soft-Delete: Second Delete Returns 200 Instead of 404

`backend/src/services/vendors.ts`

Symptom: Calling delete on a vendor that was already soft-deleted returned HTTP 200 instead of 404, making it impossible to detect whether the vendor was actually active at the time of deletion.

Root Cause: The UPDATE statement used `WHERE id = $1` without checking the current value of `is_active`. PostgreSQL still reported `rowsAffected = 1` even when setting `is_active = FALSE` on a row that was already false, because the row did match the WHERE condition.

FIX-04

Added `AND is_active = TRUE` to the WHERE clause of the soft-delete UPDATE statement. A second delete now produces `rowsAffected = 0`, which the service correctly maps to a 404 response.

BUG-05 - Inventory Matrix Test: Hardcoded Stock Assertion Fails Intermittently

`backend/src/__tests__/inventory.test.ts`

Symptom: The test "bands object contains the seeded price (299) with correct stock (20)" failed with expected 20, received 22.

Root Cause: An earlier test in the same suite added 2 additional units to the seeded batch to trigger the low-stock filter threshold condition. Because tests within a suite share the same database session, the matrix test observed the modified count of 22 rather than the original seeded value of 20.

FIX-05

Changed `expect(stockAt299).toBe(20)` to `expect(stockAt299).toBeGreaterThanOrEqual(20)`. This correctly asserts that stock is present and at least the seeded amount without making assumptions about within-suite isolation that do not hold in a shared test database.

BUG-06 - Jest 30 and ts-jest 29 Version Incompatibility

`backend/jest.config.ts` | `backend/package.json`

Symptom: All 480 tests failed immediately on startup with: Cannot find module '../app' and Cannot find module '../helpers/db'.

Root Cause: Jest 30 changed internal module resolution APIs that ts-jest 29 relied upon. This caused all relative imports in test files to fail silently. Wrong relative import paths in several test files compounded the issue.

FIX-06

Pinned Jest to version 29.7.0 which is the compatible release for ts-jest 29. Updated `jest.config.ts` to use the explicit `transform` key instead of the deprecated `globals.ts-jest` preset pattern. Corrected all relative import paths: `../../app` to `../app`, `../helpers/db` to `./helpers/db`, and fixed the schema SQL path that was one directory level off.

BUG-07 - node_modules Directory Committed to Git Repository

`.gitignore` (missing at repository root)

Symptom: The repository tracked the entire `backend/node_modules/` directory, bloating repository size by hundreds of megabytes and making all git clone and fetch operations extremely slow.

Root Cause: No `.gitignore` file existed at the repository root, so all generated and dependency directories were tracked by default from the initial commit.

FIX-07

Created a root-level `.gitignore` file excluding `node_modules/`, `dist/`, `.env`, and `*.env.local`. Ran `git rm -r --cached backend/node_modules` to untrack the committed directory without deleting local files. Repository size reduced significantly.

12. User Feedback

The following feedback was collected from the primary user interview with Bombay Fashion and from internal team review sessions. Each item is categorised and linked to the relevant user story from Milestone 1.

BUG REPORT

GPS Distance Shows Absurd Value on Clock-In Screen (Story 7.1)

Reported by: Internal testing session

Observation: The ClockWidget displayed distances such as "393191m from store" on the clock-in screen. The store GPS coordinates were set to the PostgreSQL default of latitude 0, longitude 0 which corresponds to the Gulf of Guinea off the coast of Africa. Any real device location produced a massive haversine distance against this default.

Resolution: Store coordinates were updated to the actual store location. The ClockWidget was also updated to suppress the distance indicator entirely when store coordinates are at 0, 0 so that unconfigured stores do not display misleading distance values.

BUG REPORT

GPS Enforcement Toggles Have No Effect on Clock-In (Story 7.1)

Reported by: Internal testing session

Observation: Enabling "Require GPS on Clock-In" in the Store settings page had no observable effect. Staff could clock in from any location regardless of the toggle state. The feature appeared to work visually but did not enforce anything.

Resolution: Root cause was that toggle state was stored only in the browser via Zustand and was never written to the database. Fixed by adding database columns and wiring the frontend toggles to the API. The backend attendance service now reads and enforces these flags on every clock-in and clock-out request. This is documented as BUG-01 above.

UX FEEDBACK

API Documentation Used Emojis Inconsistent with Formal Submission (Documentation)

Reported by: Team review referencing Milestone 1 PDF standards

Observation: The OpenAPI YAML description used emoji section headers which are inconsistent with the professional and formal tone of the Milestone 1 document submitted to IIT Madras.

Resolution: The entire description block in the YAML was rewritten using the exact 8 EPICs and 20 user stories from the Milestone 1 PDF in proper "As a... I want... So that..." format without any emojis. Each story maps to its corresponding API endpoint or set of endpoints.

FEATURE GAP

Reorder Message Must Be Sent Manually via WhatsApp (Story 5.2)

Reported by: Bombay Fashion interview

Observation: The system generates a well-formatted reorder message text but the manager must manually copy it and paste it into WhatsApp to send to the vendor. The store currently handles over 100 vendors and this manual step is error-prone at scale.

Status: Acknowledged and planned for Sprint 2. The Milestone 1 specification notes the use of "generic messaging or email integration instead of WhatsApp API" due to the cost and approval requirements of the WhatsApp Business API.

UX FEEDBACK

Billing Mode Switch Has No Confirmation Step (Story 3.2)

Reported by: Internal team review

Observation: Switching from Structured to Ephemeral billing mode is a consequential action since future sales will no longer be retained permanently. The current UI applies the change immediately without any confirmation dialog or warning to the owner.

Status: Planned for Sprint 2. A modal confirmation dialog with clear explanation of the consequences will be added before the mode switch is applied.

FEATURE GAP

No Quick Scan Entry for Price Band Selection During Billing (Story 3.1)

Reported by: Bombay Fashion interview

Observation: During billing, staff must scroll a dropdown list to find the correct price band. For stores with many categories this takes several taps and slows down checkout. A barcode or QR scan shortcut was requested for faster shop-floor entry.

Status: Planned for Sprint 3, subject to availability of the BarcodeDetector API on Android Chrome. Investigation will be done before committing to this feature.

PERFORMANCE

Dashboard Summary Endpoint Has No Caching (Story 8.1)

Reported by: Internal review

Observation: The GET /api/dashboard/summary endpoint runs multiple aggregation queries on every single request. On a store with 90 days of sales history the response time may degrade noticeably during peak hours when multiple staff are using the app simultaneously.

Status: A 5-minute in-process cache for dashboard query results is planned for Sprint 2.

13. Sprint 2 - Planned Improvements

The following improvements are planned for Sprint 2 based on bugs discovered during testing, user feedback received, and a review of partially implemented story acceptance criteria.

Item	Category	Story Reference	Description	Priority
Dashboard response caching	Performance	8.1, 8.2	Add a 5-minute in-process cache for GET /dashboard/summary and GET /dashboard/daily-summary to reduce repeated aggregation queries on each page load.	High
Billing mode confirmation dialog	UX Safety	3.2	Add a modal confirmation dialog before switching to Ephemeral billing mode, clearly warning the owner that future sales will not be permanently retained.	High
Reorder messaging integration	Feature	5.2	Integrate a generic email or SMS gateway so that clicking "Send Reorder" dispatches the generated message text directly to the vendor without requiring manual copy-paste into WhatsApp.	High
Automated end-of-day clock-out	Feature	7.1	Add a scheduled job at 23:59 that automatically force-clocks-out any staff member still marked as clocked in, preventing orphaned attendance records that require manual correction.	Medium
Weekly staff sales summary endpoint	Feature	7.2	Add GET /staff/summary?period=weekly returning per-staff revenue and units sold for the selected period. The current	Medium

Item	Category	Story Reference	Description	Priority
			leaderboard endpoint only covers the current calendar day.	
Discount recommendation field on aging batches	Feature	8.3	Extend the aging batch response with a suggestedDiscountPct field computed from batch age in days, cost margin percentage, and current remaining stock quantity.	Medium
Repeat purchase rate metric	Feature	6.2	Add GET /customers/repeat-rate?days=30 returning the percentage of customers who made more than one purchase within the specified window, supporting the loyalty trend visibility requirement.	Medium
Customer filter-based export	Feature	6.3	Add query parameters to GET /customers to filter by price band purchased and date range, enabling targeted contact list exports for outreach campaigns.	Medium
Automated trend insight text summaries	Feature	8.2	Add GET /dashboard/insights returning auto-generated week-over-week text summaries. Example output: "350 band sales increased 18% compared to last week."	Medium
Barcode scan shortcut for price band selection	UX	3.1	Investigate BarcodeDetector API availability on Android Chrome. If supported, add a scan button on the mobile billing screen as a faster alternative to the price band dropdown.	Low
	Security	1.1, 1.2	Enforce a limit of 10 requests per minute per IP address on	Low

Item	Category	Story Reference	Description	Priority
Rate limiting on authentication endpoints			POST /auth/login and POST /auth/register to mitigate brute-force and credential stuffing attacks.	
Pagination on high-volume list endpoints	Performance	All list endpoints	Add page and limit query parameters to GET /inventory/batches, GET /sales, and GET /customers to prevent unbounded query results and reduce memory usage on large datasets.	Low

14. Test Infrastructure

14.1 Technology Stack

Component	Technology	Version	Purpose
Test runner	Jest	29.7.0	Test execution and assertion framework
TypeScript transformer	ts-jest	29.x	Compiles TypeScript test files at runtime
HTTP integration testing	Supertest	7.x	Makes real HTTP requests to the Express app
Test database	PostgreSQL	16	Dedicated smallbiz_test database instance
PostgreSQL client	node-postgres (pg)	8.x	Database connection pool in tests
Password hashing	bcrypt	5.x	Hashes credentials for seeded test users
Runtime	Node.js	22.x	Execution environment

14.2 Database Isolation Strategy

Each test suite calls `resetDb()` in its `beforeAll` block. This re-executes `database/schema.sql` which drops all tables in reverse dependency order and recreates them from scratch. The function `seedBase()` then inserts a consistent set of fixtures for every suite: one store, one owner, one manager, one staff member, one category, one price band at price 299, one vendor, and one inventory batch of 20 units.

All test suites start from an identical, predictable database state. Tests are run serially using the `--runInBand` flag to prevent concurrent `resetDb()` calls from racing against each other on the shared test database. The npm test script is configured as

```
jest --runInBand --forceExit .
```

14.3 Test File Index

File Path	Test Count	Domain Covered
<code>src/__tests__/auth.test.ts</code>	27	Registration, login, JWT token validation
<code>src/__tests__/users.test.ts</code>	67	User CRUD, passwords, PINs, avatars, role access
<code>src/__tests__/store.test.ts</code>	41	Store profile, billing mode, retention, GPS settings
<code>src/__tests__/categories.test.ts</code>	23	Category management, uniqueness constraints
<code>src/__tests__/priceBands.test.ts</code>	32	Price band management, stock queries
<code>src/__tests__/inventory.test.ts</code>	62	Batch management, matrix view, adjust, defective, close
<code>src/__tests__/sales.test.ts</code>	22	FIFO stock deduction, customer records, custom items
<code>src/__tests__/returns.test.ts</code>	30	Returns and exchanges, stock restoration
<code>src/__tests__/vendors.test.ts</code>	45	Vendor directory, soft-delete, suggest-order logic

File Path	Test Count	Domain Covered
src/__tests__/reorders.test.ts	38	Reorder creation, status lifecycle, message text
src/__tests__/attendance.test.ts	20	Clock-in and out, GPS geofencing enforcement, roster
src/__tests__/staff.test.ts	12	Staff leaderboard, role access, sort order
src/__tests__/dashboard.test.ts	15	KPI summary, daily chart data, role access
src/__tests__/audit.test.ts	19	Audit log, ephemeral data wipe, password verification
Total	480	All 33 API endpoints covered